New games and an attempt to mimic Venema to obtain a representation-style theorem

Godhuli Mukherjee

November 10, 2025

Abstract

We introduce a new notion of games, based on a different kind of outcome relation. We then embark on the adventure of adapting Venema. Minimal necessary modifications are incorporated to carry us as far as possible without any blowback. Roadblock(s) appear too! We side-step them to resolve other areas which do not bite us back. Spoiler Alert: 'composition' turns out to be a hiccup!

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1 Set-up

We are dealing with 2-player games, as before. With any game g and each player i, we will associate an outcome relation R^i_g.

Here the outcome relation R^i_g represents the following:
If pR^i_gT holds, then from position p, the player i can reach every position in T. This is not to be confused with the outcome lying exactly in T. That's different! All we are saying is that every position in T is reachable.
So it could be true that pR^i_gU also holds when $T \subseteq U$ (just making sure every state in U is reachable would make pR^i_gU true).

An immediate natural property is the following:
(*) If $U \subseteq T$ and $pR^i_g T$, then $pR^i_g U$.(Every state in T reachable means every state in U is reachable.)

More condition(s) will arise as this discussion progresses. No spoilers!

The composed games and the meaning of outcome relations for them do not change.

2 Shifting into Algebra

As before, one can form algebras with pairs (R, S) where $R, S \in \mathcal{P}(B\ X\ \mathcal{P}(B))$ for some set B and nature of R and S are as above. Call such algebras novo-algebras. They are concretely defined!

Novo-algebras satisfy the game algebra axioms. In fact they have some more.

This is the minimal set of list that helps us replicate Venema's representation proof uninterrupted as far as possible, before one encounters the devil called composition.

$$
\begin{array}{lll}
x \vee x \approx x & x \wedge x \approx x & \text{(G1)} \\
x \vee y \approx y \vee x & x \wedge y \approx y \wedge x & \text{(G2)} \\
x \vee (y \vee z) \approx (x \vee y) \vee z & x \wedge (y \wedge z) \approx (x \wedge y) \wedge z & \text{(G3)} \\
x \vee (x \wedge y) \approx x & x \wedge (x \vee y) \approx x & \text{(G4)} \\
x \vee (y \wedge z) \approx (x \vee y) \wedge (x \vee z) & x \wedge (y \vee z) \approx (x \wedge y) \vee (x \wedge z) & \text{(G5)} \\
- - x \approx x & & \text{(G6)} \\
-(x \vee y) \approx -x \wedge -y & -(x \wedge y) \approx -x \vee -y & \text{(G7)} \\
(x \diamond y) \diamond z \approx x \diamond (y \diamond z) & & \text{(G8)} \\
(x \vee y) \diamond z \approx (x \diamond z) \vee (y \diamond z) & (x \wedge y) \diamond z \approx (x \diamond z) \wedge (y \diamond z) & \text{(G9)} \\
-x \diamond -y \approx -(x \diamond y) & & \text{(G10)} \\
y \leq z \rightarrow x \diamond y \leq x \diamond z & & \text{(G11)} \\
x \leq y \rightarrow -y \leq -x & & \text{(G12)}
\end{array}
$$

G12 is the only addition. This, ofcourse, puts more condition on the outcome relations and represents something in the nature of games. I don't know how to make all that precise right now. Let us just proceed mathematically at the moment.

A *novo-* algebra* is a structure $\mathcal{G} = (G, \vee, \wedge, -, \diamond)$ satisfying G1-G12. This is abstractly defined. A novo-algebra is a a novo-* algebra. So existence of the latter is guaranteed!

Let us introduce novo-module. A novo-module over $\mathcal{G}$ is an algebra

$$\mathcal{M} = (M, \vee, \wedge, -, \Diamond_g)_{g \in G}$$

such that $(M, \vee, \wedge, -)$ is a de Morgan algebra, and $(\Diamond_g)_{g \in G}$ is a family of unary downward monotone operators on M satisfying the following equations:

$$(M1) \quad \Diamond_{g \vee h} x \approx \Diamond_g x \vee \Diamond_h x$$

$$(M2) \quad \Diamond_{g \wedge h} x \approx \Diamond_g x \wedge \Diamond_h x$$

I don't include the next one as a necessary axiom. Including this will force the need to expand the novo-* algebra axioms to a point that doesn't even help us solve the problem of composition. So including this doesn't serve us anything. All this will be clear very very soon.

$$(M3)\ (\textit{not included}) \quad \Diamond_{g \Diamond h} x \approx \Diamond_g \Diamond_h x$$

$$(M4) \quad \Diamond_{-g} x \approx -\Diamond_g - x$$

A *novo-module* is separable if for all distinct elements g and h of G, there is an $x \in M$ such that

$$\Diamond_g x \neq \Diamond_h x.$$

Note that with this definition, we may indeed see a given novo-$*$ algebra

$$\mathcal{G} = (G, \vee, \wedge, -, \Diamond)$$

as a module over its de Morgan reduct if we put

$$\Diamond_g a = g \Diamond - a.$$

Thanks to G12 this is downward monotone. $M1$, $M2$, $M4$ are satisfied. $M3$ is satisfied if an axiom $g \Diamond - h \approx -g \Diamond h$ is added to the list of axioms G1-G12.

3 Basic Toolkit

- Let $D = (D, \vee, \wedge)$ be a distributive lattice. An ideal is a subset $I \subseteq D$ which is downward closed and closed under joins.
  An ideal I is *prime* if $a \wedge b \in I \Rightarrow a \in I$ or $b \in I$.

  Let $\mathrm{spec}(D)$ denote the set of prime ideals of D.

- Given an element $a \in D$, define

$$\hat{a} = \{p \in \mathrm{spec}(D) \mid a \in p\}.$$

Let $\mathcal{T}$ be a space of spec(D). Define

$$F_T = \{a \in D \mid \hat{a} \subseteq T\}.$$

- Let $D = (D, \vee, \wedge)$ be a distributive lattice.
A map $\diamond : D \to D$ is downward monotone iff $a \leq b \implies \diamond b \leq \diamond a$

- Let R be a binary relation on B and define the operation

$$m_R(T) = \{p \in B \mid pRT\}.$$

Easy to verify that R is downward monotone iff mR is monotone.

- Given a distributive lattice D with a downward moving map $\Diamond$, let $Q_\Diamond$ be the outcome relation on the board of prime ideals of D:

$p\, Q_\Diamond\, T$ iff $\Diamond a \in p$ for all $a \in F_T$.

Easily verified this relation is downward monotone.

- **Prime Ideal Theorem:** Let $D = (D, \vee, \wedge)$ be a distributive lattice. Let I be an ideal and $x_0 \notin I$. Then there exists a prime ideal P such that $I \subseteq P$ and $x_0 \notin P$.

- **Lemma 1:** For any distributive lattice D the map $\hat{()}$ is injective and $a \leq b \Longleftrightarrow \hat{b} \subseteq \hat{a}$
Proof: That this map is injective is proved as follows: Suppose $\widehat{a} = \widehat{b}$ but $a \neq b$. WLOG $a \ngeq b$. Consider the ideal

$$F_a = \{d \in D \mid a \geq d\}.$$

Then $a \in F_a$ but $b \notin F_a$. By the prime ideal theorem, there is some prime ideal $p \in$ spec(D) such that $a \in p$ but $b \notin p$. But this contradicts $\widehat{a} = \widehat{b}$.

To prove the map is order-reversing, let $a \leq b$, and let p, a prime ideal be in $\hat{b}$. So $b \in p$ and since p is a prime ideal, $a \in p$ by downward closure of ideals.

Now conversely, let $\hat{b} \subseteq \hat{a}$, suppose $a \nleq b$.
Consider the ideal

$$F_b = \{d \in D \mid d \leq b\}.$$

Then $a \notin F_b$ but b $\in F_b$. By Prime Ideal Theorem, there is some prime ideal P such that $F_b \subseteq P$, and $a \notin P$ but $b \in P$. Contradiction, since $\hat{b} \subseteq \hat{a}$

Lemma 2: Let D be a distributive lattice with downward monotone map $\Diamond$, then

$$pQ_\diamond\widehat{a} \text{ iff } \diamond a \in p.$$

In order to prove, first assume that $pQ_\diamond\widehat{a}$. Since $a \in F_{\widehat{a}}$ it follows that $\diamond a \in p$ by definition of $Q_\diamond$. For the other direction, suppose that $\diamond a \in p$ and let b be an arbitrary element of $F_{\widehat{a}}$. By definition of $F_{\widehat{a}}$ this means that $\widehat{b} \subseteq \widehat{a}$, so $a \leq b$. Downward monotonicity of $\diamond$ gives that $\diamond b \leq \diamond a$, so we find $\diamond b \in p$ by downward closure. Because b was arbitrary this means that $pQ_\diamond\widehat{a}$, as required.

4 Pieces of the intended proof

Assume that $M = (M, \vee, \wedge, -, \Diamond_g)_{g \in G}$ is a module over the novo-* algebra G. With every operation $\Diamond_g$ of M, we may associate a downward monotone outcome relation Q_g on *spec*(M) (for brevity, we will write Q_g rather than $Q_{\Diamond_g}$).

Proposition 1: Let $\mathcal{M} = (M, \vee, \wedge, -, \diamond_g)_{g \in G}$ be a module over $\mathcal{G} = (G, \vee, \wedge, -, \Diamond)$, let g and h be arbitrary elements of G. Then we have:

(i) $Q_{-(g \vee h)} = Q_{-g} \cup Q_{-h}$

(ii) $Q_{-(g \wedge h)} = Q_{-g} \cap Q_{-h}$

(iii) $Q_{-(g \Diamond h)} = Q_{-g} \circ Q_{-h}$ (The devil!)

(iv) If $\diamond_{-g} a \neq \diamond_{-h} a$ for some $a \in M$, then $Q_{-g} \neq Q_{-h}$.

Before discussing the (partial) proof, let us see how far this goes on to give a representation.

Proposition 2: Let $\mathcal{M} = (M, \vee, \wedge, -, \diamond_g)_{g \in G}$ be a separable module over the novo-* algebra $\mathcal{G} = (G, \vee, \wedge, -, \Diamond)$.

Then the map $\text{rep}(g) = (Q_{-g}, Q_g)$ is an embedding of G into novo-algebra on spec(M).

Recall: Novo-algebra means algebra formed with pairs (R, S) where R, S are of downward-monotone nature.

Proof:

$$\begin{aligned}\text{rep}(g \vee h) &= (Q_{-(g \vee h)}, Q_{g \vee h}) \\ &= (Q_{-(g \vee h)}, Q_{-(-g \wedge -h)}) \\ &= (Q_{-g} \cup Q_{-h}, Q_g \cap Q_h) \\ &= (Q_{-g}, Q_g) \cup (Q_{-h}, Q_h) \\ &= \text{rep}(g) \cup \text{rep}(h).\end{aligned}$$

Similarly for $g \wedge h$.

$$\text{rep}(-g) = (Q_g, Q_{-g}) = (Q_{-g}, Q_g)^- = (\text{rep}(g))^-$$

Finally, the injectivity follows from the assumed separability and part (iv) of last Proposition.

Composition stands unresolved as it is unresolved in the last theorem as well.

Proposition 3: Let $\mathcal{G} = (G, \vee, \wedge, -, \Diamond)$ be a novo-$*$ algebra. Then $\mathcal{G}$, seen as a novo module over itself, can be embedded in a separable novo-module $\mathcal{G}'$ over $\mathcal{G}$.

Proof: Almost same as Venema's original proof.

Only change is in the following:

The maps $\Diamond_g'$ in Venema's construction will now send

$$(\top, b),\ b \neq \bot \text{ to } (\bot, \bot) \quad \text{and} \quad (a, \bot),\ a \neq \top \text{ to } (\top, \top).$$

Finally (partial) proof of Proposition 1:

Proof. Let p be an arbitrary prime ideal and T an arbitrary set of prime ideals of M.

For part 1, we distinguish cases depending on the nature of T. We first assume that T is of the form $\hat{a}$ for some $a \in M$. In this case we have the following chain of equivalent statements:

$$pQ_{-(g\vee h)}\hat{a} \iff (\text{By Lemma 2})\Diamond_{-g\wedge -h}a \in p$$
$$\iff (\text{axiom of novo-module})\Diamond_{-g}a \wedge \Diamond_{-h}a \in p$$
$$\iff (p \text{ is prime})\Diamond_{-g}a \in p \text{ or } \Diamond_{-h}a \in p$$
$$\iff (\text{By Lemma 2})pQ_{-g}\hat{a} \text{ or } pQ_{-h}\hat{a}$$
$$\iff p(Q_{-g} \cup Q_{-h})\hat{a}.$$

Now allow T to be an arbitrary set. If $pQ_{-(g\vee h)}T$ then $\Diamond_{-g\wedge -h}a \in p$ for all $a \in F_T$, so for all $a \in F_T$ we have $\Diamond_{-g}a \in p$ or $\Diamond_{-h}a \in p$. We claim that $pQ_{-g}T$ or $pQ_{-h}T$, for suppose otherwise. Then there are elements $b_g, b_h \in F_T$ such that $\Diamond_{-g}b_g \notin p$ and $\Diamond_{-h}b_h \notin p$. Define $b := b_g \wedge b_h$; it is straightforward to check that $b \in F_T$. But note that since $b \leq b_g$ and $\Diamond_{-g}b_g \notin p$, we have $\Diamond_{-g}b \notin p$, otherwise we would have $\Diamond_{-g}b_g \in p$ by downward monotonicity of $\Diamond_{-g}$ and downward closure of ideal. Similarly $\Diamond_{-h}b \notin p$. Contradiction!!

For the other direction, suppose that $p(Q_{-g} \cup Q_{-h})T$; that is, we have that $pQ_{-g}T$ or $pQ_{-h}T$. Without loss of generality, suppose the first; now take any $a \in F_T$; from $pQ_{-g}T$ it follows that $\diamond_{-g}a \in p$, whence also $\diamond_{-(g\vee h)}a \in p$ since $\diamond_{-(g\vee h)}a \leq \diamond_{-g}a \in p$. Since a was an arbitrary element of F_T, this shows that $pQ_{-(g\vee h)}T$.

Part 2 of the proposition can be proved along similar lines — we omit the details.

Part 3 is unresolved.

Finally, for part 4 of the proposition, suppose that $\diamond_{-g}a \neq \diamond_{-h}a$ for some element of $\mathcal{M}$. It follows from the Prime Ideal Theorem that there is a prime ideal p that contains exactly one of these two elements, say, $\diamond_{-g}a \in p$, $\diamond_{-h}a \notin p$. It is then immediate that $pQ_{-g}\hat{a}$ but not $pQ_{-h}\hat{a}$ whence $Q_{-g} \neq Q_{-h}$.

There would arise a need for Proposition 4. It would resemble the last result in Venema's paper where a monotone outcome algebra was shown to be isomorphic to a board algebra. Here it would look like a novo-algebra being made isomorphic to an algebra which is novo with elements also having the characterisation captured by G12. This doesn't look like an easy problem to solve. If the composition problem is solved by strengthening the novo-* algebra axioms and novo-module axioms, then the final algebraic product will include more characterisations apart from the aforementioned.

5 Questions staring at us

1. What does G12 represent?
2. Can the composition part of the proof be done in this existing framework provided?
3. Can a novo-algebra be shown isomorphic to an algebra which is both novo with elements of the nature represented by G12?